

Semisimplicity of Tensor Products in Characteristic p

Pierre Deligne

English translation, literal mathematical layer

Abstract. Let G be an affine group scheme over a field k of characteristic $p \neq 0$, and let (V_i) be a finite family of semisimple representations of G . We show that, if

$$\sum_i (\dim V_i - 1) < p,$$

then the tensor-product representation of G on $\bigotimes_i V_i$ is again semisimple. Under the additional hypothesis that G is smooth, this theorem was proved by J.-P. Serre in 1994. We reduce to that case. More generally, the V_i may be taken to be objects of a Tannakian category over k .

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1 Introduction

Let T be a Tannakian category over a field k of characteristic $p \neq 0$. For the definition of Tannakian categories we refer to [3], p. 193, or to [1], 2.1 and 2.8. Recall that among the axioms are the existence of duals, and the existence of a fiber functor ω with values in the vector spaces over a suitable extension K of k . The dimension $\dim(V)$ of an object V of T is $\dim_K \omega(V)$. Since two fiber functors become isomorphic over a suitable larger extension ([1] 1.12, 1.13), this dimension does not depend on the choice of the fiber functor ω .

Theorem 0.1. Let $(V_i)_{i \in I}$ be a finite family of objects of T . If the V_i are semisimple and

$$\sum_{i \in I} (\dim(V_i) - 1) < p,$$

then the tensor product $\bigotimes_{i \in I} V_i$ is a semisimple object of T .

Example 0.2. If T is the category of finite-dimensional linear representations of an affine group scheme over k , one obtains the theorem announced in the abstract.

Other examples are the category of finite-dimensional representations of a Lie algebra over k , or that of finite-dimensional p -representations of a p -Lie algebra over k . By passage to an algebraic envelope, these cases also reduce to the preceding one.

Example 0.3. Suppose that a group Γ acts on a commutative field K of characteristic p , and let $k := K^\Gamma$ be the subfield of invariants. Let T be the category of finite-dimensional K -vector spaces, endowed with a semilinear action of Γ . With the tensor product over K , this category is Tannakian over k : it admits the fiber functor over K “underlying vector space.” If semilinear representations V_i of Γ on K are semisimple and

$$\sum_i (\dim_K(V_i) - 1) < p,$$

then the semilinear tensor-product representation of the V_i is therefore semisimple.

0.4. Let V be a finite-dimensional vector space over k of characteristic p . As explained in [4], §4, an automorphism g of V such that $g^p = 1$ defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V), \quad t \longmapsto g^t,$$

where g^t is defined by the binomial formula

$$g^t := \sum_{n < p} \binom{t}{n} (g - 1)^n. \quad (0.4.1)$$

Similarly, an endomorphism N of V such that $N^p = 0$ defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V), \quad t \longmapsto \exp(tN),$$

where $\exp(tN)$ is defined by

$$\exp(tN) := \sum_{n < p} \frac{(tN)^n}{n!}. \quad (0.4.2)$$

Indeed, after any extension of scalars to a commutative k -algebra Λ , if for $t \in \Lambda$ one defines $\exp(tN)$ by (0.4.2), the hypothesis $N^p = 0$ ensures that

$$\exp((t' + t'')N) = \exp(t'N) \exp(t''N).$$

Definition 0.5. Suppose that k is algebraically closed of characteristic p , and let G be a subgroup scheme of $\mathrm{GL}(V)$.

- (i) G is *saturated* if, for every $g \in G(k)$ such that $g^p = 1$, the morphism $t \mapsto g^t$ from $\mathbb{G}_a$ to $\mathrm{GL}(V)$ factors through G .
- (ii) G is *infinitesimally saturated* if, for every $N \in \mathrm{Lie}(G)$ such that $N^p = 0$ (in the sense of the p -Lie-algebra structure of $\mathrm{Lie}(G)$, or as an endomorphism of V , which amounts to the same thing), the morphism $t \mapsto \exp(tN)$ from $\mathbb{G}_a$ to $\mathrm{GL}(V)$ factors through G .

We shall say that G is *doubly saturated* if it is saturated and infinitesimally saturated.

0.6. Let G be an affine group scheme of finite type over an algebraically closed field k of characteristic p . In the statement 0.1, take T to be the category of finite-dimensional representations of G , suppose that the family (V_i) of representations is faithful, and put $V := \bigoplus V_i$. To prove 0.1 when G is reduced, Serre [4] begins by reducing to the case in which G is saturated in $\mathrm{GL}(V)$. If that is the case, the finite group $\pi_0(G)$ of connected components of G has order prime to p , and this permits a reduction to the case in which G is connected. For G not necessarily reduced, we use an analogous argument to reduce to the case in which G is infinitesimally saturated in $\mathrm{GL}(V)$, and our key result in that case is the following structure theorem. This theorem will provide a reduction to the case in which G is smooth.

Let G^0 be the identity component of G , let G_{red}^0 be the reduced subgroup scheme of G^0 , and let $R_{\mathrm{u}} G_{\mathrm{red}}^0$ be the unipotent radical of G_{red}^0 . Both G_{red}^0 and $R_{\mathrm{u}} G_{\mathrm{red}}^0$ are smooth, i.e. linear algebraic groups in the sense of the Chevalley seminar of 1956/58.

Theorem 0.7. Let $(V_i)_{i \in I}$ be a finite faithful family of representations of G . Let V be the sum of the V_i . Suppose that each representation V_i has dimension $< p$ and that, as a subgroup of $\mathrm{GL}(V)$, G is infinitesimally saturated. Under these hypotheses:

- (i) G_{red}^0 and $R_{\mathrm{u}} G_{\mathrm{red}}^0$ are invariant subgroup schemes of G , and G^0/G_{red}^0 is of multiplicative type.
- (ii) If the representations V_i are semisimple, G_{red}^0 is reductive.
- (iii) If G_{red}^0 is reductive, there exists in G^0 a central connected subgroup scheme M of multiplicative type such that the morphism

$$M \times G_{\mathrm{red}}^0 \longrightarrow G^0$$

realizes G^0 as a quotient of $M \times G_{\mathrm{red}}^0$.

2 Proof of Theorem 0.7

Proposition 1.1. In an affine group scheme G of finite type over a field, every connected subgroup scheme of multiplicative type is contained in a maximal connected subgroup scheme of multiplicative type.

Proof, after a letter from M. Raynaud to the author. It is enough to show that, in the set of connected subgroup schemes of multiplicative type of G , ordered by inclusion, every totally ordered subset $\mathcal{M}$ is bounded above (Bourbaki, Ens. III, §2, no. 4, Theorem 2). For M in $\mathcal{M}$, the centralizer $Z_G(M)$ of M in G is a subgroup scheme, automatically closed, and the map $M \mapsto Z_G(M)$ is decreasing. The ordered set of closed subgroup schemes of G is noetherian. The set of the $Z_G(M)$ for M in $\mathcal{M}$ therefore has a least element C . Every M in $\mathcal{M}$ is contained in the center $Z(C)$ of C . This center is affine and commutative. The identity component of its largest subgroup scheme of multiplicative type (SGA3 XVII 7.2.1) bounds $\mathcal{M}$ from above. This is the required upper bound.

1.2. Fix k , G , and the finite family of representations $(V_i)_{i \in I}$, with sum V , satisfying the hypotheses of 0.7: k is algebraically closed of characteristic $p \neq 0$, $\dim(V_i) < p$, and, as a subgroup scheme of $\mathrm{GL}(V)$, G is infinitesimally saturated.

Lemma 1.3.

- (i) The identity component G^0 of G is infinitesimally saturated in $\mathrm{GL}(V)$.
- (ii) If a representation W of G is semisimple, its restriction to G^0 is also semisimple.
- (iii) If the reduced subgroup G_{red}^0 of G^0 (respectively its unipotent radical $R_{\mathrm{u}} G_{\mathrm{red}}^0$) is an invariant subgroup scheme of G^0 , then it is one of G .

Proof. Assertion (i) follows from $\mathrm{Lie} G = \mathrm{Lie} G^0$ and from the connectedness of $\mathbb{G}_a$.

It is enough to prove (ii) for W irreducible. Let $S \subset W$ be the sum of the irreducible G^0 -subrepresentations of W . We must show that $S = W$. Each $g \in G(k)$ normalizes G^0 . By transport of structure, S is therefore stable under g . Since k is algebraically closed, G is the union of the gG^0 for $g \in G(k)$, and S is stable under G . Since W is irreducible and $S \neq 0$, S is all of W .

Similarly, by transport of structure, each $g \in G(k)$ normalizes G_{red}^0 and $R_{\mathrm{u}} G_{\mathrm{red}}^0$, and (iii) follows because G is the union of the gG^0 .

1.4. Lemma 1.3 shows that, to prove 0.7 for G , it suffices to prove it for G^0 . Moreover, if $p = 2$, each V_i has dimension one, G is of multiplicative type, and 0.7 is trivial.

In the rest of this section, besides the hypotheses of 1.2, we shall suppose that $p \neq 2$ and that G is connected: $G = G^0$. We shall prove that G_{red} is an invariant subgroup scheme of G in 1.18, that G/G_{red} is of multiplicative type in 1.19, the invariance in G of $R_u G_{\text{red}}$ in 1.20–1.23, 0.7(ii) in 1.24, and 0.7(iii) in 1.25.

1.5. Let H be a maximal connected subgroup of multiplicative type of G (1.1), and let $Z_G^0(H)$ be the identity component of its centralizer $Z_G(H)$. The quotient

$$U := Z_G^0(H)/H$$

contains no subgroup μ_p , since the inverse image in $Z_G^0(H)$ of such a subgroup would be an extension of μ_p by H , hence of multiplicative type (SGA3 XVII 7.1.1 b)). Thus U is unipotent (SGA3 XVII 4.3.1(v)).

1.6. Let X be the group of characters of H . The connectedness of H is equivalent to X having no torsion prime to p . Giving an action of H on a vector space W is equivalent to giving an X -grading

$$W = \bigoplus_{\alpha \in X} W^\alpha$$

of W , and this construction is functorial and compatible with tensor product. In particular, V and the V_i are X -graded, and the action by inner automorphisms of H on G defines an X -grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha$$

of $\text{Lie}(G)$.

Lemma 1.7. The central extension $Z_G^0(H)$ of U by H is split.

Since U is unipotent and H is of multiplicative type, there is no nontrivial morphism from U to H (SGA3 XVII 2.4(ii)). The splitting guaranteed by Lemma 1.7 is therefore unique. We shall use the corresponding isomorphism of $H \times U$ with $Z_G^0(H)$ to regard U as a subgroup of $Z_G^0(H)$.

Proof. Let V_i^α ($\alpha \in X$) be the homogeneous components of the V_i . The group $Z_G^0(H)$ acts on V_i^α . Let χ_i^α be the determinant character of this representation. Its restriction to H is $\dim(V_i^\alpha)$ times the character α of H .

Let $J \subset I \times X$ be the set of pairs (i, α) such that $V_i^\alpha \neq 0$, and for $j = (i, \alpha)$ put

$$V^j := V_i^\alpha, \quad \chi_j := \chi_i^\alpha.$$

The family (V_i) of representations of H is faithful: the α for $j = (i, \alpha)$ in J generate X . If $j = (i, \alpha)$ is in J , $\dim V_i^\alpha \leq \dim V_i < p$ is prime to p . The restrictions to H of the χ_j ($j \in J$) therefore generate a subgroup of X of finite index prime to p . Let F be the kernel of

$$(\chi_j) : H \longrightarrow \mathbb{G}_m^J$$

and let Q be the cokernel. The group F is finite, reduced, and of order prime to p . In the commutative diagram with exact rows

$$\begin{array}{ccccccc} 1 & \longrightarrow & H/F & \longrightarrow & Z_G^0(H)/F & \longrightarrow & U \longrightarrow 1 \\ & & \parallel & & \downarrow & & \downarrow \\ 1 & \longrightarrow & H/F & \longrightarrow & \mathbb{G}_m^J & \longrightarrow & Q \longrightarrow 1, \end{array}$$

the morphism from the unipotent group U to the group of multiplicative type Q is trivial. The kernel of (χ_j) therefore lifts U into $Z_G^0(H)/F$. Its inverse image in $Z_G^0(H)$ is a central extension E of U by F . By Lemma 1.8 below, this extension is trivial. A lifting of U into E splits the central extension $Z_G^0(H)$ of U by H .

Lemma 1.8. A central extension of a unipotent group by a finite reduced group of order prime to p is trivial.

We give two proofs of the lemma, based on different ideas. The second assumes that U is connected, which is sufficient for the application above.

First proof. Let

$$1 \longrightarrow F \longrightarrow E \longrightarrow U \longrightarrow 1$$

be a central extension as in the lemma, and let a be an integer prime to p that kills F . View E as a central extension in the category of fppf sheaves of groups on $\text{Spec}(k)$. For any S/k , $U(S)$ admits a finite central filtration with successive quotients killed by p (apply SGA3 XV 3.5(v)). Hence every finite subset X of $U(S)$ generates a finite p -group P_X . Let Q_X be its intersection with the image of $E(S)$. The inverse image of Q_X in $E(S)$ is a central extension E_X of Q_X by $F(S) = F^{\pi_0(S)}$. The group $H^2(Q_X, F(S))$ classifying these extensions is trivial, since it is killed both by a and by the power $|Q_X|$ of p . The extension E_X is therefore trivial. Moreover, the lifting of Q_X into E_X is unique: it is the set of the a -th powers of elements of E_X .

Passing to the inductive limit over X and sheafifying, one deduces that the subsheaf of sets of E which is the image of

$$x \longmapsto x^a : E \longrightarrow E$$

is a subsheaf in groups lifting U into E .

Second proof. The action by right translations of F on E makes E an F -torsor over U . It is trivialized over the identity element, and therefore corresponds to a morphism

$$\pi_1(U, e) \longrightarrow F.$$

The reduced scheme U_{red} is isomorphic to an affine space $\mathbb{A}^N$. Since

$$\pi_1(U, e) = \pi_1(U_{\text{red}}, e) \simeq \pi_1(\mathbb{A}^N, 0)$$

has no nontrivial quotient of order prime to p (SGA4 XV 2.2), the F -torsor E is trivial: the identity component of E lifts U into E .

Lemma 1.9. Let $f : V \rightarrow W$ be a morphism of schemes of finite type over k , let $v \in V(k)$, and let $(df)_v$ be the morphism induced by f from the Zariski tangent space of V at v to the Zariski tangent space of W at $f(v)$. If V is smooth over k and $(df)_v$ is an isomorphism, then f is etale at v and W is smooth over k in a neighborhood of $f(v)$.

Proof. Let A (respectively B) be the completion of the local ring of V at v (respectively of W at $f(v)$). Let m be the maximal ideal of A and n that of B . The injectivity of $(df)_v$ ensures that A is a quotient of B . We must show that $B \xrightarrow{\sim} A$ (SGA1 I 4.4). Let $t_1, \dots, t_N$ in n lift a basis of n/n^2 . The bijectivity of $(df)_v$ ensures that the images of the t_i in m/m^2 form a basis. The t_i define an epimorphism

$$g : T_i \mapsto t_i : k[[T_1, \dots, T_N]] \longrightarrow B.$$

In the commutative diagram

$$\begin{array}{ccc} & k[[T_1, \dots, T_N]] & \\ \swarrow & & \searrow g \\ A & \xleftarrow{\hspace{1.5cm}} & B \end{array}$$

the smoothness of V at v ensures that g is an isomorphism. It follows that $B \xrightarrow{\sim} A$.

In characteristic p , the Lie algebra of a group scheme is a p -Lie algebra. We shall write $\gamma \mapsto \gamma^p$ for the p -th power operation, often denoted $\gamma^{[p]}$. For every linear representation ρ of the group, one has

$$\rho(\gamma^p) = \rho(\gamma)^p.$$

Lemma 1.10. For every $\gamma \in \text{Lie } U$, one has $\gamma^p = 0$.

As explained under 1.7, we regard here U as a subgroup of $Z_G^0(H) \subset G$.

Proof. Since U is a unipotent group scheme, for each i the image γ_i of γ in $\text{End}(V_i)$ is nilpotent. Since $\dim(V_i) < p$, one has $\gamma_i^p = 0$. Since the family (V_i) of representations is faithful, $\gamma^p = 0$.

Corollary 1.11. The unipotent group U is smooth.

Proof. Let $(\gamma_a)_{1 \leq a \leq A}$ be a basis of $\text{Lie } U$. By 1.10, for each a ,

$$\rho_a : \lambda \mapsto \exp(\lambda \gamma_a) : \mathbb{G}_a \longrightarrow \text{GL}(V)$$

is defined by 0.4. The hypothesis that G is infinitesimally saturated ensures that this morphism factors through G . Since U centralizes H , the action of H by inner automorphisms fixes $\text{Lie } U$, hence each ρ_a , and ρ_a factors through $Z_G(H)$, and even through $Z_G^0(H) = H \times U$ because $\mathbb{G}_a$ is connected. Every morphism from $\mathbb{G}_a$ to H being trivial, ρ_a factors through U . Let $\mathbb{E}^A$ be affine space of dimension A and set

$$f : \mathbb{E}^A \longrightarrow U, (t_a) \mapsto \prod \rho_a(t_a),$$

the product being taken with respect to the group law of U . By construction, df is an isomorphism at 0. Applying 1.9, one deduces that U is smooth in a neighborhood of the identity element, hence everywhere by homogeneity.

Recall that the action by inner automorphisms of H on $\text{Lie } G$ defines a grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha.$$

Lemma 1.12. If $\alpha \in X$ is nonzero and $\gamma \in \text{Lie}(G)^\alpha$, then $\gamma^p = 0$.

Proof. Regard γ as an endomorphism of V . If γ^p were nonzero, in the sense of the p -Lie algebra structure of $\text{Lie } G$ or as an endomorphism of V (which amounts to the same thing), there would exist $i, \beta \in X$, and $v \in V_i^\beta$ such that $\gamma^p v \neq 0$. This implies the nonvanishing of the $\gamma^l v$ for $0 \leq l \leq p$. The element $\gamma^l v$ of V_i lies in $V_i^{\beta + l\alpha}$. For $0 \leq l < p$, the $\beta + l\alpha$ are distinct, because X has no torsion prime to p . The $\gamma^l v$ for $0 \leq l < p$ are therefore linearly independent. This contradicts the hypothesis $\dim V_i < p$.

1.13. Let $(\gamma_b)_{1 \leq b \leq B}$ be a family of elements of $\text{Lie } G$, the disjoint union of bases of the $(\text{Lie } G)^\alpha$ for $\alpha \neq 0$. Let α_b be the character α of H such that $\gamma_b \in (\text{Lie } G)^\alpha$. Since G is infinitesimally saturated, the morphism $\lambda \mapsto \exp(\lambda \gamma_b) : \mathbb{G}_a \rightarrow \text{GL}(V)$ provided by 0.4 and 1.12 factors through a morphism

$$\rho_b : \mathbb{G}_a \longrightarrow G.$$

This morphism is equivariant for the action of H on $\mathbb{G}_a$ through α_b and on G by inner automorphisms, because the same equivariance holds after composition with the inclusion of G in $\text{GL}(V)$.

Let q (respectively q_0) be the projection of G onto $G/Z_G(H)$ (respectively $G/Z_G^0(H)$). Consider the morphism

$$f : \mathbb{E}^B \longrightarrow G, \quad (t_b) \longmapsto \prod_{b=1}^B \rho_b(t_b),$$

the product being taken with respect to the group law of G .

Lemma 1.14. The composite morphism $qf : \mathbb{E}^B \rightarrow G/Z_G(H)$ is étale at 0.

Since $G/Z_G^0(H)$ is étale over $G/Z_G(H)$, the same statement holds for q_0f .

Proof. Let e be the identity element of G , and let $T_{q(e)}$ denote the tangent space of $G/Z_G(H)$ at $q(e)$. By 1.9, it is enough to check that qf induces an isomorphism from the tangent space at 0 onto $T_{q(e)}$, and for that it is enough to check that

$$dq : \text{Lie}(G) \longrightarrow T_{q(e)}$$

induces an isomorphism

$$\bigoplus_{\alpha \neq 0} \text{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

If G were smooth, $Z_G(H)$ would be smooth as the fixed locus of an action on G of the group of multiplicative type H ; one would deduce an exact sequence

$$0 \longrightarrow \text{Lie } Z_G^0(H) \longrightarrow \text{Lie } G \longrightarrow T_{q(e)} \longrightarrow 0,$$

and one would conclude by noting that $\text{Lie } Z_G^0(H) = \text{Lie}(G)^H$.

We shall deduce 1.14 from the following lemma.

Lemma 1.15. Suppose that a group H of multiplicative type, with character group X , acts on an affine k -scheme of finite type $S = \text{Spec}(A)$. Let S^H be the fixed locus and let $s \in S^H(k)$. Let I be the ideal defining S^H in S . View I/I^2 as a coherent sheaf on S^H , and let

$$(I/I^2)_s = s^*(I/I^2)$$

be its fiber at s . Let m be the maximal ideal at s . The group H acts on m/m^2 , hence m/m^2 is X -graded. The inclusion of I in m induces an isomorphism

$$(I/I^2)_s \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (m/m^2)^\alpha. \quad (1.15.1)$$

The hypothesis “ S affine” is probably unnecessary.

Proof. The action of H gives an X -grading of A . The quotient A/I is the largest graded quotient of A purely of degree 0: I is the ideal generated by the A^α for $\alpha \neq 0$. The two sides of (1.15.1) are unchanged when A is replaced by A/I^2 . We may therefore suppose that $I^2 = 0$. Then $A^\alpha A^\beta = 0$ for $\alpha, \beta \neq 0$, I is the sum of the A^α for $\alpha \neq 0$, and $A^0 \xrightarrow{\sim} A/I$. Let m_0 be the maximal ideal of s in $A/I \xleftarrow{\sim} A^0$. We have

$$m = m_0 \oplus \bigoplus_{\alpha \neq 0} A^\alpha.$$

Thus

$$m/m^2 = m_0/m_0^2 \oplus \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/m_0,$$

and

$$I/I^2 \otimes_{A^0} A^0/m_0 = \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/m_0,$$

which gives (1.15.1).

End of the proof of 1.14. The morphism $G \rightarrow G/Z_G(H)$ is flat. If I is the ideal defining $Z_G(H)$ in G , and n is the maximal ideal at the point $q(e)$ of $G/Z_G(H)$, then the vector bundle I/I^2 on $Z_G(H)$ is the inverse image of n/n^2 on $q(e)$. Its fiber at the identity element e of G is n/n^2 , and therefore by (1.15.1)

$$n/n^2 \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (m/m^2)^\alpha.$$

Passing to duals gives

$$\bigoplus_{\alpha \neq 0} \mathrm{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

Corollary 1.16. The morphism

$$F : \mathbb{E}^B \times Z_G^0(H) \longrightarrow G, \quad (t_b), z \longmapsto \prod \rho_b(t_b) \cdot z$$

is etale at $(0, e)$.

Proof. The square

$$\begin{array}{ccc} \mathbb{E}^B \times Z_G(H) & \xrightarrow{F} & G \\ \downarrow & & \downarrow \\ \mathbb{E}^B & \xrightarrow{q_0 f} & G/Z_G^0(H) \end{array}$$

is cartesian. Hence F is etale at every point lying above $0 \in \mathbb{E}^B$.

1.17. If $f : T \rightarrow U$ is a quasi-compact morphism of schemes, the closed scheme-theoretic image $\overline{f(T)}$ of f is the smallest closed subscheme of U through which f factors. If $f : \mathrm{Spec}(A) \rightarrow \mathrm{Spec}(B)$ is affine, it is defined by the ideal kernel of $f^* : B \rightarrow A$. Its formation is compatible with every flat base change $U' \rightarrow U$. If f is a morphism of schemes over S , if a group scheme H flat over S acts on T and U , and if f is equivariant, this compatibility with base change implies that the action of H on U induces an action of H on the closed scheme-theoretic image $\overline{f(T)}$.

1.18. Proof of the invariance of G_{red} in G . The morphism F of 1.16 is equivariant for the action of H on $\mathbb{E}^B$ by the α_b , and on G by inner automorphisms. Since U is smooth (1.11), the reduced subscheme of

$$Z_G^0(H) = H \times U$$

is $H_{\text{red}} \times U$. An étale morphism induces an étale morphism on reduced subschemes. The morphism induced by F from $\mathbb{E}^B \times H_{\text{red}} \times U$ to G_{red} is therefore étale at 0, and its image contains a nonempty open of G_{red} . The closed scheme-theoretic image of the morphism

$$F_1 : \mathbb{E}^B \times H_{\text{red}} \times U \longrightarrow G$$

induced by F therefore coincides with G_{red} . Since this morphism is H -equivariant, 1.17 ensures that G_{red} is stable under the action of H on G by inner automorphisms.

Because F is étale at the origin, G is generated as an fppf sheaf by the image of F , hence by H and G_{red} , since $U \subset G_{\text{red}}$. Since H normalizes G_{red} , it follows that G_{red} is an invariant subgroup scheme of G .

Proposition 1.19. One has

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}.$$

Consequently G/G_{red} is of multiplicative type.

Proof. If Y_1 and Y_2 are two closed subschemes of Z , then for any morphism $u : Z' \rightarrow Z$ one has

$$u^{-1}(Y_1 \cap Y_2) = u^{-1}(Y_1) \cap u^{-1}(Y_2).$$

If u is étale, and $Y_2 = Z_{\text{red}}$, one obtains

$$u^{-1}(Y_1 \cap Z_{\text{red}}) = u^{-1}(Y_1) \cap Z'_{\text{red}}.$$

Apply this to F , which is étale in a neighborhood of $\{0\} \times H \subset \{0\} \times Z_G^0(H)$. One obtains that, in a neighborhood of $\{0\} \times H$, the inverse image of $H \cap G_{\text{red}}$ is H_{red} ; hence $H_{\text{red}} \rightarrow H \cap G_{\text{red}}$ is étale, therefore an isomorphism. If H , H_{red} , G , G_{red} and the quotients are interpreted as fppf sheaves, the isomorphism

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}$$

follows from H normalizing G_{red} , from $HG_{\text{red}} = G$, and from $H_{\text{red}} \xrightarrow{\sim} H \cap G_{\text{red}}$.

1.20. Proof of the invariance in G of $R_{\text{u}} G_{\text{red}}$. Let R denote the unipotent radical $R_{\text{u}} G_{\text{red}}$ of the smooth group G_{red} , and let

$$L := G_{\text{red}}/R.$$

Let T_L be a maximal torus of the reductive group L . It is a maximal connected subgroup of multiplicative type. The inverse image of T_L in G_{red} is an extension of the torus T_L by the smooth connected unipotent group R . By SGA3 XVII 5.1.1(i), such an extension is trivial: the torus T_L lifts to a torus T , automatically maximal, of G_{red} . We shall take H to be a maximal connected subgroup of multiplicative type of G containing T . Then

$$T = H_{\text{red}} = H \cap G_{\text{red}}.$$

Lemma 1.21. The connected centralizer $Z_{G_{\text{red}}}^0(T)$ of T in G_{red} is a product $T \times W$, with W unipotent. The group scheme H normalizes W .

Proof. Since H normalizes T , H normalizes

$$Z := Z_{G_{\text{red}}}^0(T).$$

As with G , the subgroup G_{red} of $\text{GL}(V)$ is infinitesimally saturated. Applying 1.7 to G_{red} and T (instead of G and H) gives the decomposition

$$Z = T \times W.$$

The proof of 1.7 describes $W \subset Z$ as the identity component of the intersection of the kernels of the determinant characters of the representations of Z in the V_i^β , for β a weight of T in V . This description shows that H normalizes W .

Lemma 1.22. The action of H on $\text{Lie}(G_{\text{red}})$ preserves the subspace $\text{Lie}(R)$.

Proof. Decompose

$$l = \text{Lie } L, \quad g = \text{Lie } G_{\text{red}}, \quad r = \text{Lie } R$$

according to the weights of T . Since H normalizes T , the decomposition obtained for g is respected by H . We shall show that each r^α , hence their sum r , is stable under H .

The connected centralizer of T_L in L is reduced to T_L . Hence we have a morphism of exact sequences

$$\begin{array}{ccccccc} 1 & \longrightarrow & W & \longrightarrow & Z = T \times W & \longrightarrow & T_L \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & R & \longrightarrow & G_{\text{red}} & \longrightarrow & L. \end{array}$$

The part of weight 0 of g is $\text{Lie}(Z)$. Its intersection with $\text{Lie } R$ is $\text{Lie } W$, stable under H by 1.21.

If α is not a weight of T on l , then $r^\alpha = g^\alpha$, which is stable under H .

If α is a nonzero weight of T on l , then l^α and $l^{-\alpha}$ have dimension one. Under our hypothesis that $p \neq 2$, the bracket induces a nondegenerate pairing of l^α and $l^{-\alpha}$ with values in a line $h_\alpha \subset \text{Lie } T_L$. The bracket of g induces a pairing of g^α and $g^{-\alpha}$ with values in

$$g^0 = \text{Lie } Z.$$

Its composite with $g^0 \rightarrow g^0 / \text{Lie } W \simeq \text{Lie } T_L$ factors through $g^{\pm\alpha} \rightarrow l^{\pm\alpha}$ and the preceding pairing. It therefore has values in a line d of $g^0 / \text{Lie } W$, and

$$r^{\pm\alpha} = \text{Ker}(g^{\pm\alpha} \rightarrow \text{Hom}(g^{\mp\alpha}, d)) :$$

the $r^{\pm\alpha}$ are the kernels of the pairing of g^α and $g^{-\alpha}$ with values in d . The decomposition of g according to the weights of T , and $W \subset g^0$, are respected by H ; this description shows that the r^α , and hence r , are subrepresentations of H acting on g .

1.23. End of the proof of the invariance of R . Let Fil denote the finite increasing filtration of $\bigoplus V_i$ such that $\text{Fil}_0 = 0$ and $\text{Fil}_{n+1} / \text{Fil}_n$ is the space of invariants of R in $\bigoplus V_i / \text{Fil}_n$. It is the sum of filtrations of the V_i . It is respected by G_{red} , since R is an invariant subgroup of G_{red} .

The reduced subgroup of the largest subgroup of G_{red} which acts trivially on

$$\text{Gr}^{\text{Fil}}(\bigoplus V_i)$$

is a smooth invariant unipotent subgroup containing R . It therefore coincides with R .

If $\gamma \in \text{Lie } R$, then γ acts nilpotently on V_i . Since the family of representations V_i of dimension $< p$ is faithful, one has $\gamma^p = 0$. Since G_{red} is infinitesimally saturated, the corresponding morphism

$$\rho : \mathbb{G}_a \longrightarrow \prod \text{GL}(V_i)$$

factors through G_{red} . Since this representation of $\mathbb{G}_a$ is trivial on

$$\text{Gr}^{\text{Fil}}(\bigoplus V_i),$$

the morphism ρ even factors through R .

Choose a basis $(\gamma_c)_{1 \leq c \leq C}$ of $r = \text{Lie } R$, the disjoint union of bases of the r^β for β a weight of H acting on r . Let $\rho_c : \mathbb{G}_a \rightarrow R$ be the morphism defined by γ_c . As in 5.6, the morphism of schemes

$$f : \mathbb{E}^C \longrightarrow R, \quad (t_c) \longmapsto \prod_{c=1}^C \rho_c(t_c)$$

is étale at 0. Its composite with the inclusion of R in G_{red} is H -equivariant, for a suitable action of H on $\mathbb{E}^C$. By the same argument as in 1.17 and 1.18, it follows that R is stable under the action of H on G_{red} .

Corollary 1.24. If the representations V_i are semisimple, the group scheme G_{red} is reductive.

Proof. Replacing each V_i by a family of irreducible representations of which V_i is the sum, we reduce to supposing the V_i irreducible. We must prove that the unipotent radical $R = R_u G_{\text{red}}$ is trivial.

Since R is unipotent, the subspace V_i^R of R -invariants in V_i is nonzero. Because R is invariant in G , V_i^R is stable under G , hence, by irreducibility of V_i , coincides with V_i . Since the family of the V_i is faithful, R is trivial.

1.25. Proof of 0.7(iii). Suppose that G_{red} is reductive, let T be a maximal torus of G_{red} , and take H to be a maximal connected subgroup of multiplicative type of G containing T . It normalizes G_{red} and its action on G_{red} fixes T . The group scheme of automorphisms of the reductive group G_{red} which fix T is the image T^{ad} of T in the adjoint group. If Z is the subgroup of H which centralizes G_{red} , the morphism

$$Z \times T \longrightarrow H$$

is a quotient by T : the image T^{ad} of T in the adjoint group. View all these group schemes as fppf sheaves. Since H and G_{red} generate G , Z is central. Since the action of H on G_{red} factors through a quotient of T , Z and T generate H . Since Z is generated by Z^0 and Z_{red} and $Z_{\text{red}} \subset G_{\text{red}}$, Z^0 and G_{red} generate G : the morphism

$$Z^0 \times G_{\text{red}} \longrightarrow G$$

is an epimorphism, and this verifies 0.7(iii).

3 Saturation (cf. [4] §4)

2.1. Let V be a finite-dimensional vector space over an algebraically closed field k of characteristic p , and let G be a subgroup scheme of $\text{GL}(V)$. The intersection G^* of the doubly saturated subgroup schemes (0.5) of $\text{GL}(V)$ containing G is doubly saturated. We call G^* the double saturation of G .

Let $(V_i)_{i \in I}$ be a finite family of finite-dimensional vector spaces over k . For each i , let N_i be an endomorphism of V_i such that $N_i^p = 0$. It defines an action $t \mapsto \exp(tN_i)$ of $\mathbb{G}_a$ on V_i (0.4). For $a \in I$, again denote by N_a the endomorphism of $\bigotimes V_i$ which is the tensor product of $N_a \in \text{End}(V_a)$ and of the identities $1 \in \text{End}(V_j)$ for $j \neq a$. Let N be the endomorphism $\sum N_a$ of $\bigotimes V_i$.

Proposition 2.2. Suppose there exist integers v_i such that $N_i^{v_i} = 0$ and

$$\sum_i (v_i - 1) < p.$$

Then $N^p = 0$, and the action $t \mapsto \exp(tN)$ of $\mathbb{G}_a$ on $\bigotimes V_i$ is the tensor product of the actions of $\mathbb{G}_a$ on the V_i defined by the N_i .

Proof. The endomorphisms N_a of $\bigotimes V_i$ commute with one another, and every monomial of degree p in the N_a has degree at least v_a in N_a for some $a \in I$, hence is zero. It follows that

$$N^p = \left(\sum N_a \right)^p = 0,$$

and also that the truncated exponentials satisfy

$$\exp(tN) = \exp\left(\sum tN_a\right) = \prod \exp(tN_a) = \bigotimes \exp(tN_a).$$

Corollary 2.3. If $\sum_i (\dim(V_i) - 1) < p$, the action $\exp(tN)$ of $\mathbb{G}_a$ on $\bigotimes V_i$ is the tensor product of the actions of $\mathbb{G}_a$ on the V_i defined by the N_i .

Proof. Take $v_i = \dim(V_i)$ in 2.2.

By the same arguments as those of [4], §4, one deduces from [4], §4, Proposition 10, and from 2.3 the following proposition 2.4. Let $(V_i)_{i \in I}$ be a finite family of finite-dimensional vector spaces over k and let G be a subgroup scheme of $\prod \text{GL}(V_i)$. The subgroup scheme $\prod \text{GL}(V_i)$ of $\text{GL}(\bigoplus V_i)$ is saturated and infinitesimally saturated. The double saturation G^* of G , viewed as a subgroup of $\text{GL}(\bigoplus V_i)$, is therefore contained in $\prod \text{GL}(V_i)$.

Proposition 2.4. If $\sum_i (\dim(V_i) - 1) < p$, then a subspace W of $\bigotimes V_i$ is stable under the action of G if and only if it is stable under that of G^* . Thus the representation $\bigotimes V_i$ of G is semisimple if and only if it is semisimple as a representation of G^* .

4 The case in which k is algebraically closed

Let G be an affine group scheme over an algebraically closed field k of characteristic p , and let $(V_i)_{i \in I}$ be a finite family of representations of G .

Proposition 3.1. If the representations V_i are semisimple and $\sum_i (\dim V_i - 1) < p$, then the representation $\bigotimes V_i$ of G is again semisimple.

Proof. We leave to the reader the verification that this statement is implied by the special case in which one makes the additional hypotheses that each V_i is simple of dimension ≥ 2 , and that $|I| \geq 2$. These hypotheses imply that $p \neq 2$, and that each V_i has dimension $< p$.

Replacing G by its image in $\prod \text{GL}(V_i)$, one may suppose that G is a subgroup scheme of $\prod \text{GL}(V_i)$. By 2.4, one may moreover suppose that, as a subgroup of $\text{GL}(\bigoplus V_i)$, G is doubly saturated. In the rest of the proof we shall suppose all these properties satisfied. That G is

saturated is equivalent to G_{red} being so and, by [4], §4, Proposition 11, implies that the finite group

$$G/G^0 = G_{\text{red}}/G_{\text{red}}^0$$

has order prime to p .

By 0.7(ii)(iii), the group G_{red}^0 is reductive and G^0 contains a central connected group M of multiplicative type such that

$$M \times G_{\text{red}}^0 \longrightarrow G^0$$

is an epimorphism.

By Lemma 1.3(ii), the V_i are semisimple as representations of G^0 . If W is an irreducible representation of G^0 , then M , being of multiplicative type and central, acts on W by a character. The representation W is therefore still irreducible as a representation of G_{red}^0 . Thus the V_i are semisimple as representations of G_{red}^0 , and by [4] the tensor product of these representations is again semisimple as a representation of G_{red}^0 . The following lemma shows that it is semisimple as a representation of G .

Lemma 3.2. Suppose that a group scheme G is an extension of a linearly reductive group scheme A by a group scheme B . Then every representation V of G which is semisimple as a representation of B is semisimple.

That A is linearly reductive means that all its linear representations are semisimple. A group of multiplicative type, and a finite group of order prime to p , are linearly reductive.

Proof. Let S be the sum of the irreducible subrepresentations of V . We must show that $S = V$, and to verify this it is enough to show that S has a complement T in V stable under G . Consider the exact sequence

$$1 \longrightarrow \text{Hom}_B(V/S, S) \longrightarrow \text{Hom}_B(V, S) \longrightarrow \text{Hom}_B(S, S) \longrightarrow 1.$$

It is an exact sequence of representations of A . Since A is linearly reductive, the identity 1_S of S , fixed by A , lifts to a homomorphism $f : V \rightarrow S$ of representations of B which is fixed by A , i.e. is a morphism of representations of G . Its kernel is the required complement.

Corollary 3.3. The statement 0.1 is true when k is algebraically closed.

Proof. Let T' be the Tannakian subcategory of T generated from the V_i by tensor product, duality, and passage to a subquotient. By [3] 3.3.1.1, T' admits a fiber functor over k , hence is equivalent to the category of representations of an affine group scheme over k . Another reference is [1] 6.20.

Remark 3.4. The reduction from T to T' is in fact unnecessary. I have checked (unpublished) that every Tannakian category over an algebraically closed field k is equivalent to the Tannakian category of representations of an affine group scheme over k .

5 Extension of scalars in Tannakian categories and proof of 0.1

4.1. Let k be a commutative field and let T be a k -linear abelian category. Define the tensor product of an object X of T and a finite-dimensional k -vector space V to be the object of T which represents the functor

$$Y \longmapsto V \otimes \text{Hom}(Y, X).$$

The choice of a basis $(e_i)_{i \in I}$ of V identifies $V \otimes X$ with the sum of a family of copies of X indexed by I .

Let k' be a finite extension of k . We define as follows the k' -linear abelian category $T_{k'}$ obtained from T by extension of scalars to k' , the scalar-extension functor

$$e_{k'/k} : T_k \longrightarrow T_{k'},$$

and the scalar-restriction functor

$$r_{k \setminus k'} : T_{k'} \longrightarrow T_k.$$

The category $T_{k'}$ is the category of objects X of T endowed with a k' -module structure $k' \otimes X \rightarrow X$. For X in T ,

$$e_{k'/k}(X) := k' \otimes X,$$

endowed with its natural k' -module structure. The functor $r_{k \setminus k'}$ forgets the k' -module structure. The functors $e_{k'/k}$ and $r_{k \setminus k'}$ are exact, and $r_{k \setminus k'}$ is right adjoint to $e_{k'/k}$. Example: if T is the category of left modules over a k -algebra A , then $T_{k'}$ identifies with the category of left modules over the k' -algebra

$$A' := k' \otimes_k A,$$

and $e_{k'/k}$ and $r_{k \setminus k'}$ are the usual extension- and restriction-of-scalars functors.

Suppose that the objects of T have finite length and that the Hom groups are finite-dimensional over k . If X is a simple object of T , the division algebra $\text{End}(X)$, being finite-dimensional over k , is central simple over its center, which is a finite extension of k .

Lemma 4.2. Under these hypotheses:

- (i) Let X be a simple object of T , $D := \text{End}(X)$, and F the center of D . The object $e_{k'/k}(X)$ of $T_{k'}$ is semisimple if and only if $F \otimes_k k'$ is a product of fields.
- (ii) If Y is a semisimple object of $T_{k'}$, then $r_{k \setminus k'}(Y)$ is semisimple.
- (iii) If X is in T and $e_{k'/k}(X)$ is semisimple, then X is semisimple.

Proof. (i) The functor $Z \mapsto \text{Hom}(X, Z)$ is an equivalence from the category of objects of T which are sums of copies of X to the category of finite-dimensional right D -vector spaces. The category of sums of copies of X endowed with a k' -module structure identifies with the category of finite-type right $D \otimes_k k'$ -modules. The object $e_{k'/k}(X)$ of $T_{k'}$ thus corresponds to the $D \otimes_k k'$ -module $D \otimes_k k'$. It is known to be semisimple if and only if $F \otimes_k k'$ is a product of fields.

(ii) Reduce to the case in which Y is simple. The sum S of the simple subobjects of $r_{k \setminus k'}(Y)$ is nonzero and is a k' -submodule. By the simplicity of Y , it coincides with $r_{k \setminus k'}(Y)$.

(iii) The object X is a direct factor of $r_{k \setminus k'} e_{k'/k}(X)$, and one applies (ii).

4.3. Now suppose that T is a Tannakian category over k , and equip $T_{k'}$ with the tensor product over k' :

$$X \otimes_{k'} Y = \text{coker}(X \otimes k' \otimes Y \rightrightarrows X \otimes Y).$$

The scalar-extension functor $e_{k'/k}$ is compatible with tensor products and with their associativity and commutativity data. It sends the unit object to the unit object, and duals, in the sense of [1] 2.2, to duals. The category $T_{k'}$ admits internal Homs: if X and Y in T are endowed with k' -module structures, their internal Hom in $T_{k'}$ is the kernel of the two morphisms

$$\text{Hom}(X, Y) \rightrightarrows \text{Hom}(k' \otimes X, Y) = k'^{\vee} \otimes \text{Hom}(X, Y),$$

where $k'^{\vee}$ is the k -dual of the k -vector space k' . We shall denote it by $\text{Hom}_{k'}(X, Y)$.

Theorem 4.4. If T is a Tannakian category over k and k' is a finite extension of k , then the category $T_{k'}$, endowed with the tensor product over k' , is Tannakian over k' .

It is possible to deduce this theorem from Theorem 1.12 of [1], which characterizes Tannakian categories as categories of representations of affine groupoids G acting transitively on a k -scheme S ($G \rightarrow S \times_k S$ affine and faithfully flat): if T is such a category of representations, then $T_{k'}$ identifies with the category of representations of the groupoid obtained from it by extension of scalars to k' .

Conversely, and according to a strategy indicated to me by Grothendieck about thirty years ago, the more elementary proof of 4.4 that follows should make it possible to prove [1] 1.12 more simply.

If $k' \supset k'' \supset k$, then $T_{k'}$ identifies with $(T_{k''})_{k'}$. This observation permits one to reduce the proof of 4.4 to the following two special cases: k'/k separable, and k'/k inseparable of degree p . The delicate point is to show the existence of duals, in the sense of [1] 2.2, in $T_{k'}$. Once this is acquired, if ω is a fiber functor over the k -algebra K , then ω provides a fiber functor ω' of $T_{k'}$ over the k' -algebra

$$K' := k' \otimes_k K :$$

to X in T , endowed with a k' -module structure, attach $\omega(X)$, endowed with its k' -module structure. This structure makes $\omega(X)$ a $k' \otimes K$ -module.

Proof if k' is a finite separable extension of k . If X is in T and has $X^\vee$ as dual, in the sense of [1] 2.2, the object $e_{k'/k}(X)$ of $T_{k'}$ admits $e_{k'/k}(X^\vee)$ as dual. One concludes by observing that every object Y of $T_{k'}$ is a direct factor of an object of this form. More precisely, the adjunction morphism

$$e_{k'/k} r_{k \setminus k'}(Y) \longrightarrow Y$$

is split: if one regards Y as an object of T , $k' \otimes Y$ has two natural k' -module structures, one coming from k' (this is that of $e_{k'/k} r_{k \setminus k'}(Y)$), the other from Y . The adjunction morphism is the natural morphism

$$k' \otimes Y \longrightarrow (k' \otimes Y) \otimes_{k' \otimes k} k'.$$

It is split because the morphism $k' \otimes_k k' \rightarrow k'$ is the projection onto a direct factor.

Proof if $k' = k(a^{1/p})$. Let ω be a fiber functor over an extension K of k . The key point is the following lemma.

Lemma 4.5. If an object X of T is endowed with a k' -module structure, this structure makes $\omega(X)$ a $k' \otimes_k K$ -module. This module is free.

Proof. If $K' := k' \otimes_k K$ is a field, this is clear. Otherwise, a has a p -th root α in K , and the K -algebra K' is isomorphic to $K[\varepsilon]/(\varepsilon^p)$: take

$$\varepsilon = \sqrt[p]{a} - \alpha.$$

Let d be the dimension over K of $\omega(X)/\varepsilon\omega(X)$. The K' -module $\omega(X)$ is isomorphic to a sum of d monogenic modules $K[\varepsilon]/(\varepsilon^j)$, $1 \leq j \leq p$. One checks that it is free if and only if the exterior power over K'

$$\bigwedge \text{olimits}_{K'}^d \omega(X)$$

is free of rank one. By the exactness of ω , this exterior power over K' is the image under ω of the exterior power over k' of X , defined as the image of the antisymmetrization

$$a : \bigotimes \text{olimits}_{k'}^d X \longrightarrow \bigotimes \text{olimits}_{k'}^d X.$$

Replacing X by this exterior power, we are reduced to the case in which X is such that $\omega(X)$ is a nonzero monogenic K' -module.

The k' -module structure of X gives a morphism

$$k' \otimes \mathbf{1} \longrightarrow \text{Hom}(X, X). \quad (4.5.1)$$

One knows that the unit object $\mathbf{1}$ is simple ([2] 1.17). The kernel of (4.5.1) is therefore of the form $A \otimes \mathbf{1}$, for A a vector subspace of k' . This subspace is a proper ideal of k' , so $A = 0$ and (4.5.1) is a monomorphism. Applying ω , one deduces that $\omega(X)$ is a faithful K' -module, and one concludes by observing that a faithful monogenic K' -module is free.

4.6. End of the proof if $k' = k(a^{1/p})$. Let $\mathbf{1}'$ denote the unit object $e_{k'/k}(\mathbf{1})$ of $T_{k'}$. For X in $T_{k'}$, put

$$X^\vee := \text{Hom}_{k'}(X, \mathbf{1}').$$

We show that $X^\vee$ is a dual of X in the sense of [1] 2.2. In $T_{k'}$ there is an evaluation morphism

$$\text{ev} : X^\vee \otimes_{k'} X \longrightarrow \mathbf{1}' \quad (4.6.1)$$

and, for every Y in $T_{k'}$, a morphism

$$Y \otimes_{k'} X^\vee \longrightarrow \text{Hom}_{k'}(X, Y). \quad (4.6.2)$$

The functor ω' , with values in K' -modules, transforms this morphism into

$$\omega(Y) \otimes \omega(X)^\vee \longrightarrow \text{Hom}_{K'}(\omega(X), \omega(Y)). \quad (4.6.3)$$

Lemma 4.5 ensures that (4.6.3) is an isomorphism. Since ω' is exact and satisfies $\omega(Z) = 0 \Rightarrow Z = 0$, because ω has these properties, it is conservative and (4.6.2) is an isomorphism. Taking $Y = X$, and composing the evident morphism $\mathbf{1}' \rightarrow \text{Hom}_{k'}(X, X)$ with the inverse of (4.6.2), we obtain

$$\delta : \mathbf{1}' \longrightarrow X \otimes_{k'} X^\vee.$$

The morphisms ev and δ satisfy the identities [1] (2.1.2) making $X^\vee$ a dual of X , since these identities become true after application of ω' .

Corollary 4.7. Let X be an object of a Tannakian category T over k . If X admits a module structure over an extension k' of k , then $[k' : k]$ divides $\dim X$.

Proof. Let ω be a fiber functor of T over an extension K of k , put $K' := k' \otimes K$, and regard $\omega(X)$, with its k' -module structure deduced from that of X , as a K' -module. If X is regarded as an object of the Tannakian category $T_{k'}$, this K' -module is the image of X under the fiber functor ω' of $T_{k'}$ deduced from ω . As the image of the object X of the Tannakian category $T_{k'}$ under the fiber functor ω' , it is a locally free K' -module of constant rank, equal to the dimension of X viewed as an object of $T_{k'}$. Thus

$$\begin{aligned} \dim(X) &:= \dim_K \omega(X) \\ &= [K' : K] \cdot \text{rank}_{K'} \omega(X) \\ &= [k' : k] \cdot \dim(X \text{ viewed as an object of } T_{k'}). \end{aligned}$$

4.8. Warning. If D is a division algebra over k and a nonzero object X admits a D -module structure, one does not necessarily have $\dim X \geq [D : k]$. Example: take T to be the category of $\mathbb{Z}/2$ -graded complex vector spaces V endowed with a homogeneous antilinear automorphism j whose square is 1 on V^0 and -1 on V^1 . Tensor product is taken over $\mathbb{C}$. This category is Tannakian over $\mathbb{R}$, and over $\mathbb{C}$ admits the fiber functor “underlying complex vector space.” An odd object identifies with a vector space over the quaternion field $\mathbb{H}$. A simple odd object S has dimension 2 and $\text{End}(S)$ is isomorphic to $\mathbb{H}$.

Corollary 4.9. If X is a simple object of a Tannakian category T over k of characteristic p and $\dim X < p$, then the center of $\text{End}(X)$ is a separable extension of k .

Proof. By 4.7 it is an extension of degree $< p$.

4.10. Let k_1 be an algebraic extension of k . If T is a k -linear abelian category, the categories $T_{k'}$, for k' a finite extension of k in k_1 , form a 2-inductive system of abelian categories. I write “2-inductive” rather than “inductive,” because the scalar-extension functors

$$e_{k''/k'} : T_{k'} \longrightarrow (T_{k'})_{k''} = T_{k''}$$

for $k' \subset k''$ satisfy

$$e_{k'''/k''} e_{k''/k'} = e_{k'''/k'}$$

only up to a canonical isomorphism; these isomorphisms satisfy a cocycle condition for $k' \subset k'' \subset k''' \subset k''''$. Define T_{k_1} to be the 2-inductive limit of the $T_{k'}$, for k' a finite extension of k in k_1 . Since the functors $e_{k''/k'}$ are exact, the category T_{k_1} is again abelian.

If T is Tannakian, with fiber functor ω over K , the categories $T_{k'}$ for $k' \subset k_1$ are Tannakian, and ω provides a fiber functor of $T_{k'}$ over $k' \otimes K$. Passing to the inductive limit, one sees that T_{k_1} is Tannakian, that ω provides a fiber functor ω_1 of T_{k_1} over $k_1 \otimes K$, and that the $e_{k'/k}$ provide a scalar-extension functor

$$e_{k_1/k} : T \longrightarrow T_{k_1},$$

compatible with tensor product.

Corollary 4.11. Let X be an object of T .

- (i) If the object $e_{k_1/k}(X)$ of T_{k_1} is semisimple, then X is semisimple.
- (ii) If $\dim(X) < p$ and X is semisimple, then $e_{k_1/k}(X)$ is semisimple.

Proof. For k' a finite extension of k in k_1 , the length of $e_{k'/k}(X)$ increases with k' and is bounded by $\dim(X)$. There exists therefore a finite extension k' such that the length of $e_{k''/k}(X)$ is constant for $k'' \supset k'$.

Prove (i) for X . By 4.2(iii), it is enough to show that for a suitable finite extension k'' of k' in k_1 , $e_{k''/k'}(X)$ is semisimple. Let

$$0 = X_0 \subset X_1 \subset X_2 \subset \cdots \subset X_n = e_{k'/k}(X)$$

be a filtration of $e_{k'/k}(X)$ with simple successive quotients S_i . The extension of S_i by X_{i-1} defining X_i becomes split over k_1 . There is therefore a finite extension k'' of k' in k_1 over which these extensions split. The S_i remain simple because

$$\lg e_{k''/k}(X) = \lg e_{k'/k}(X).$$

It follows that $e_{k''/k'}(X)$ is semisimple.

For (ii), let X be semisimple with $\dim(X) < p$. It is enough to treat X simple. By 4.9, the center of $\text{End}(X)$ is separable over k ; hence, by 4.2(i), all finite scalar extensions occurring in the construction preserve semisimplicity, and so $e_{k_1/k}(X)$ is semisimple.

4.12. Proof of 0.1. Let T be a Tannakian category over k of characteristic p , and take for k_1 an algebraic closure $\bar{k}$ of k . Corollary 4.11 reduces 0.1 for T to 0.1 for $T_{\bar{k}}$, proved in 3.3.

6 Tensor products of exterior powers

The results that follow were suggested by the referee.

Theorem 0.7 makes it possible to generalize to objects of a Tannakian category Serre's theorem [5] on the semisimplicity of tensor products of exterior powers of representations.

Theorem 5.1. Let T be a Tannakian category over a field k of characteristic $p > 0$, let $(V_i)_{i \in I}$ be a finite family of semisimple objects of T , and let $(m_i)_{i \in I}$ be a family of integers ≥ 0 . If

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.1.1)$$

then the tensor product of the $\bigwedge^{m_i} V_i$ is semisimple.

Recall that the m -th exterior power of an object V of T is the image of the antisymmetrization

$$a : \bigotimes \text{olimits}^m V \longrightarrow \bigotimes \text{olimits}^m V.$$

For every fiber functor ω , one has

$$\omega(\bigwedge \text{olimits}^m V) = \bigwedge \text{olimits}^m \omega(V),$$

and the dual of $\bigwedge^m V$ is $\bigwedge^m (V^\vee)$.

We shall reduce the proof of 5.1 to the special case in which, for each i , $\dim(V_i) < p$. The arguments of 3.3, 4.11 and 4.12 then reduce further to supposing that k is algebraically closed, that T is the category of representations of an affine group scheme G over k , and that the family of representations (V_i) is faithful, i.e. that G is a subgroup scheme of $\prod \text{GL}(V_i)$, and hence of $\text{GL}(V)$, for $V := \bigoplus V_i$.

After a few preliminaries (5.3 to 5.10), we shall reduce in 5.11 to the case in which G is doubly saturated in $\text{GL}(V)$, and treat that case in 5.12.

5.2. Reduction to the case $\dim(V_i) < p$. We successively reduce to supposing that:

- (1) $m_i \leq \dim V_i$; otherwise $\bigwedge^{m_i} V_i = 0$.
- (2) $0 < m_i < \dim V_i$; otherwise $\bigwedge^{m_i} V_i$ has dimension one and the factor $\bigwedge^{m_i} V_i$ can be omitted. This changes neither hypothesis (5.1.1) nor the conclusion, since if L has dimension one, semisimplicity of X is equivalent to that of $X \otimes L$, the lattices of subobjects of X and $X \otimes L$ being isomorphic.
- (3) $m_i \leq (\dim V_i)/2$; otherwise replace $\bigwedge^{m_i} V_i$ by $\bigwedge^{\dim V_i - m_i} V_i^\vee$, which differs from it only by tensoring with the rank-one object $\bigwedge^{\dim V_i} V_i$.
- (4) $\sum m_i \geq 2$; otherwise the tensor product is 1 (if $\sum m_i = 0$) or is reduced to a V_i (if $\sum m_i = 1$).

If these conditions are satisfied, then $\dim V_i < p$. Indeed, if $|I| \geq 2$, (5.1.1) implies

$$\dim V_i - 1 = 1(\dim V_i - 1) \leq m_i(\dim V_i - m_i) < p - 1,$$

and if $|I| = 1$, then $m_i \geq 2$, $\dim V_i \geq 4$, and

$$\dim V_i \leq 2(\dim V_i - 2) \leq m_i(\dim V_i - m_i) < p.$$

5.3. Let V be a representation of $\mathrm{SL}(2)$. The action of the multiplicative group of diagonal matrices (a, a^{-1}) defines a grading

$$V = \bigoplus V^j$$

of V . The action of

$$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SL}(2)$$

exchanges V^j and V^{-j} . Hence $\dim V^j = \dim V^{-j}$. Let E be the action of the element

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

of the Lie algebra. This endomorphism of V has degree 2. If the weights of V are $< v$, i.e. if $V^j = 0$ for $|j| \geq v$, then $E^v = 0$.

If the weights of V are $< p$, one knows that the representation V is semisimple, a sum of Sym^i of the fundamental representation with $i < p$, and that the j -th iterate E^j of E induces an isomorphism

$$E^j : V^{-j} \xrightarrow{\sim} V^j. \quad (5.3.1)$$

Conversely, if V is a graded vector space, with $V^j = 0$ for $|j| \geq p$, and if E is an endomorphism of degree 2 of V satisfying (5.3.1), then there exists a unique action of $\mathrm{SL}(2)$ on V defining this grading and this endomorphism; one has $E^p = 0$, and the action of

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$$

in $\mathrm{SL}(2)$ is given by the truncated exponential

$$\exp(tE) := \sum_{n < p} \frac{(tE)^n}{n!}. \quad (5.3.2)$$

5.4. Let E be a nilpotent endomorphism of a vector space V of dimension $\leq p$. Decomposing E into Jordan blocks, one checks that E comes from a representation of $\mathrm{SL}(2)$ as in 5.3, with weights $< p$. If E has only one Jordan block, the dominant weight is $\dim(V) - 1$.

Lemma 5.5. With the hypotheses and notation of 5.4, the weights of the representation $\bigwedge^m V$ of $\mathrm{SL}(2)$ are $\leq m(\dim V - m)$.

Lifting the representation V of $\mathrm{SL}(2)$ to characteristic 0, one reduces to checking in characteristic 0 that, for a representation W of $\mathrm{SL}(2)$ of dimension d , one has

$$\text{the weights of the representation } \rho_m \text{ of } \mathrm{SL}(2) \text{ on } \bigwedge^m W \text{ are } \leq m(d - m). \quad (5.5.1)$$

Since, in characteristic 0, every representation of $\mathrm{SL}(2)$ is a direct sum of Sym^j of the fundamental representation, (5.5.1) is equivalent to

$$(d\rho_m \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix})^{m(d-m)+1} = 0. \quad (5.5.2)$$

Let E be the nilpotent endomorphism of W giving the action of $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$. If one identifies endomorphisms with actions of the one-dimensional Lie algebra, then

$$d\rho_m \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

is the natural extension E_m of E to $\bigwedge^m W$. Nilpotent endomorphisms with a single Jordan block are dense among nilpotent endomorphisms. Hence it is enough to check that

$$(E_m)^{m(d-m)+1} = 0$$

when E has a single Jordan block. In that case the weights of the representation W are

$$(d-1), (d-3), \dots, -(d-1),$$

each with multiplicity one. The largest weight of the representation $\bigwedge^m V$ is therefore

$$(d-1) + (d-3) + \dots + (d-(2m-1)) = md - m^2 = m(d-m),$$

and (5.5.2) follows.

5.6. Return to characteristic p . Let N be a nilpotent endomorphism of a vector space V of dimension $d \leq p$, and let N_m be its extension, in the sense of Lie algebras as in 5.5, to $\bigwedge^m V$. Since $d \leq p$, one has $N^p = 0$ and N defines an action

$$\rho(t) = \exp(tN)$$

of $\mathbb{G}_a$ on V . This action extends to an action of $\mathbb{G}_a$ on $\bigwedge^m V$.

Proposition 5.7. With the notation of 5.6, one has

$$(N_m)^{m(d-m)+1} = 0, \tag{5.7.1}$$

and, if $m(d-m) < p$, the action of $\mathbb{G}_a$ on $\bigwedge^m V$ is given by the truncated exponential $\exp(tN_m)$.

Proof. As in 5.4, there exists an action ρ of $\mathrm{SL}(2)$ on V , with weights $< p$, such that

$$d\rho \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = N.$$

It extends to an action ρ_m of $\mathrm{SL}(2)$ on $\bigwedge^m V$. By 5.5, $\bigwedge^m V$ has weights $\leq m(d-m)$. The vanishing (5.7.1) follows.

If $m(d-m) < p$, then by 5.4 the action of

$$\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}(2)$$

on $\bigwedge^m V$ is given by the truncated exponential $\exp(tN_m)$.

5.8. Let (V_i) be a finite family of vector spaces of dimension $< p$, and let (m_i) be a family of integers ≥ 0 . For each i , let N_i be a nilpotent endomorphism of V_i . The truncated exponential $\exp(tN_i)$ is an action of $\mathbb{G}_a$ on V_i . These actions define an action of $\mathbb{G}_a$ on

$$\bigotimes_i \bigwedge^{m_i} V_i$$

and, by passage to the Lie algebra, an endomorphism N of this tensor product. It is deduced from the N_i , viewed as actions on the V_i of the one-dimensional Lie algebra.

Corollary 5.9. With the notation of 5.8, if

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.9.1)$$

then $N^p = 0$ and the action of $\mathbb{G}_a$ on $\bigotimes_i \bigwedge^{m_i} V_i$ is given by the truncated exponential $\exp(tN)$.

Proof. As in 5.4, choose an action ρ_i of $\mathrm{SL}(2)$ on V_i , with weights $< p$, such that

$$d\rho_i \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = N_i.$$

By 5.5, the weights of $\mathrm{SL}(2)$ acting on $\bigotimes_i \bigwedge^{m_i} V_i$ are

$$\leq \sum_i m_i(\dim V_i - m_i) < p,$$

and one applies 5.4 as in 5.7. One could equally well apply 5.7 and 2.2.

5.10. The truncated exponential induces a bijection between nilpotent endomorphisms N with $N^p = 0$ and automorphisms g with $g^p = 1$, and the corresponding actions of $\mathbb{G}_a$ coincide:

$$\exp(N)^t = \exp(tN).$$

With the notation of 5.8, if the automorphism g_i of V_i satisfies $g_i^p = 1$, and if g is the automorphism of $\bigotimes_i \bigwedge^{m_i} V_i$ induced by the g_i , then condition (5.9.1) ensures that g^t is induced by the g_i^t .

Recall that, by the remarks following 5.1, 5.2 reduces the proof of 5.1 to the case in which k is algebraically closed and T is the category of representations of

$$G \subset \prod \mathrm{GL}(V_i) \subset \mathrm{GL}(V),$$

with $V := \bigoplus_i V_i$.

5.11. Reduction to the case in which G is doubly saturated in $\mathrm{GL}(V)$. Let G^* be the double saturation of G (2.1). As in 2.4, 5.9 and 5.10 ensure that, if (5.1.1) is satisfied, a subspace of $\bigotimes_i \bigwedge^{m_i} V_i$ stable under G is automatically stable under G^* . Semisimplicity as a representation of G^* therefore implies semisimplicity as a representation of G .

5.12. End of the proof of 5.1. As in 3.1, one reduces to supposing G reduced. One can then apply [5], p. 26, consequence of the “Main Theorem” on p. 20, to the representations V_i of $G(k)$. One could instead invoke [6].

5.13. In 5.1, we use the Schur functors

$$V_i \mapsto \bigwedge^{m_i} V_i,$$

associated with the representations of $\mathrm{GL}(\dim V_i)$ of dominant weight

$$(1, \dots, 1, 0, \dots, 0)$$

(m_i entries equal to 1). More generally one can consider Schur functors $S_{\lambda(i)}$, where $\lambda(i)$ is a dominant weight

$$(\lambda(i)_1, \dots, \lambda(i)_{\dim V_i})$$

of $\mathrm{GL}(\dim V_i)$. Regard $\lambda(i)$ as a partition. Let $\mu(i)$ be its dual and put

$$n(\lambda(i)) = \sum_j \mu(i)_j (\dim V_i - \mu(i)_j)$$

(cf. [6] 5.2). Condition (5.5.1) is to be replaced by

$$\sum_i n(\lambda(i)) < p. \quad (5.13.1)$$

That (5.13.1) and the semisimplicity of the V_i suffice to ensure the semisimplicity of

$$\bigotimes_i S_{\lambda(i)} V_i$$

can be deduced from 5.1 and from the fact that $S_{\lambda(i)}$ is a subquotient, and even, under the hypotheses made, a direct factor, of the functor

$$V \mapsto \bigotimes_j \bigwedge^{\lambda(i)_j} V$$

for V of dimension $\dim(V_i)$.

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References

- [1] P. Deligne, *Catégories tannakiennes*, in *The Grothendieck Festschrift*, vol. 2, pp. 111–194. Progress in Math. 87, Birkhäuser, 1990.
- [2] P. Deligne and J. S. Milne, *Tannakian categories*, in *Hodge Cycles, Motives and Shimura Varieties*, by P. Deligne, J. S. Milne, A. Ogus and K. Shih, pp. 101–228. Springer Lecture Notes in Mathematics 900.
- [3] N. Saavedra, *Catégories Tannakiennes*, Springer Lecture Notes in Mathematics 265.
- [4] J.-P. Serre, *Sur la semi-simplicité des produits tensoriels de représentations de groupes*, Invent. Math. 116 (1994), pp. 513–530.
- [5] J.-P. Serre, *Moursund Lectures 1998*, notes by W. E. Duckworth, available on arXiv: math/0305257.
- [6] J.-P. Serre, *Complète réductibilité*, Sémin. Bourbaki 2003–2004, exp. 932, in Astérisque 299 (Soc. Math. France, 2005).

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SGA3: Séminaire 1962/64, directed by M. Demazure and A. Grothendieck: *Schémas en Groupes*. The exposé XVII of Michel Raynaud is in volume II, Springer Lecture Notes in Mathematics 152.

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